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Games

Alexander Nevo

Alexander Nevo
3h

In volatility calibration,
Too much rigidity can s
microstructure noise or

At its core, calibration i

Two broad philosophies
comparison focuses on
approximations):

1 Rigid Prior Geometry
Classical parametric m
stable surfaces.

Viewed geometrically, calibration becomes a constrained deformation of a prior surface family. The admissible manifold itself acts as a filter, restricting allowable deformations and preventing “free lunch”.

Under rigid priors, localized perturbations may cease to remain admissible. The observed quote then appears as a geometric breach relative to the constrained manifold.

2 Flexible Prior Geometry (e.g. Arbitrage-Constrained Deep Learning)

Neural approaches relax geometric restrictions while preserving no-arbitrage structure through penalties or soft constraints.

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 Inference-sensitive anomalies (solid circles)

Rigid priors may classify these as geometric breaches, while flexible priors may accommodate them through localized deformation.

 Prior-sensitive local

However, flexibility comes at the cost of neighboring distortions.

 Local adaptivity can induce structures. This raises the question of whether

 One emerging direction is to use a backbone while learning

Related ideas appear in [Surfaces with SABR-informed learned market adaptivity](#). SABR approximation with exact implied volatilities

The broader application of active research directions

age may induce

uding rough/jagged
erely fitted noise?

analytical geometry as
ns.

s for Implied Volatility
SABR priors with
combine Hagan's
against numerically

remains an open and

81 · 4 Comments

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Donn Bryan Julian CLSSBB

1h

Is the flexible geometry learning market structure, or just learning which quote errors the rigid prior refused to absorb?

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Patricio Galvan

2h

Very interesting and insightful post, Alexander. I'd be curious to look a bit deeper into the implementation side. Do you happen to have a GitHub repo or any code examples you could

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